

history over the past 250 years: the transformation from an agricultural society that relied on windmills, watermills, horses, and other beasts of burden, to an industrial one, in which steam engines were capable of generating reliable power on a scale that was previously unimaginable.

The background

The industrialization process that started in Britain around the mid- to late 18th century was initiated by the scientific revolution in Europe in the late 17th century. Of similar importance were financial changes pioneered in the Netherlands, then imported to Britain: more readily available credit helped boost entrepreneurial activities. It had never been easier for members of the increasingly wealthy middle class, looking for ways to invest their money, to support new inventions and technologies.

A third factor was an agricultural revolution, which began in the Netherlands and Britain, where farmers realized that crop rotation made it unnecessary to leave land fallow every third year. In both of these countries, land reclamation increased the acreage available for farming. Crop yields were thus boosted, just as selective breeding produced larger and more profitable domesticated animals—sources of food and wool alike. With any likelihood of famine now receding, the population of Britain grew, between 1750 and 1800, from 6.5 million to over 9 million. This, in turn, meant new markets and an expanded workforce.

In Britain, an improved transportation network allowed goods, produced on an ever-larger scale, to be transported faster and more reliably. Between 1760 and 1800, as many as 4,250 miles (6,840km) of canals were built in England.

Thinkers sought to understand the impulses behind these societal changes. The publication in 1776 of *The Wealth of Nations* by the Scottish philosopher Adam Smith underpinned what was becoming known as political economy, and the central role of the profit motive and of competition in increasing efficiencies and lowering prices.

This economic transformation also contributed to and was, in turn, boosted

by the emergence of global markets—a consequence of burgeoning European colonial empires, which offered greater access to raw materials and also provided markets for finished goods. A better-mapped world, and improvements in ship types and position-finding at sea, also facilitated global trade.

Steam power

The overriding force behind the economic transformation, though, was the development of the steam engine. In an astonishingly short time, it would revolutionize Britain, making it the world's first industrial power, and ultimately transform the world. Even so, it might never have had its dramatic global impact had Britain not had huge reserves of the fuel needed to make it work: coal. The replacement of wood as the prime source of fuel was critical to industrial development. In exactly the same way, the development of coke (processed coal that burns at much higher temperatures than coal) at the beginning of the 18th century would make the production of iron—the indispensable core material of the new technologies—faster and simpler.

Steam engines of varying degrees of reliability had been developed as early as 1712, when Thomas Newcomen built an “atmospheric engine.” But it was only with James Watt's first rotating steam engine in 1781 that the extraordinary potential of machine power became clear. The earliest steam engines had been used mainly as pumps. Watt's rotating engine, on the other hand, could power machinery. The engineering company he and Matthew Boulton established in Birmingham in 1775 produced over 500 steam engines.

When Watt's patents expired in 1800, others started producing their own steam engines. The textile industries in the northwest benefited in particular from the increased availability of steam power, and large-scale, almost entirely mechanized, factory production soon replaced small, home-based manufacturing. By 1835, there were more than 120,000 power looms in textile mills. No longer dependent on rivers as power sources, factories could be built »